

**Q1 : What are the advantages and disadvantages of using a convex mirror for seeing traffic at the rear?
(CBSE 2015)**

Advantage of Convex Mirror: It provide wider view of the rear traffic.

Disadvantage: It does not give the correct distance of the vehicle at the rear.

13. What is the radius of curvature of a plane mirror?

Ans : [CBSE 2015]

The radius of curvature of a plane mirror is infinite.

20. How can you identify the three types of mirrors without touching them?

Ans : [CBSE 2011]

- a. If the image formed is erect, of same size, and at equal distance behind the mirror as object and is in front of the mirror, then mirror is plane.
- b. If the image is of larger size and erect and changes to inverted and smaller size by changing the position of mirror from the object it is concave mirror.
- c. If the image is smaller and erect for all positions of mirror from the object, it is a convex mirror.

22. (a) What should be the position of the object, when a concave mirror is to be used:
- (i) as a shaving mirror, and
 - (ii) in torches producing parallel beam of light?
- (b) A man standing in front of a mirror finds his image having a very small head and legs of normal size. What type of mirror are used in designing such a mirror?

Ans : [CBSE 2011]

- (a) (i) Face must be placed in between the pole and focus of the mirror so that enlarged, erect image of face can be formed.
- (ii) At focus, because rays coming from the focus after reflection will become parallel.
- (b) It is a combination of convex mirror (small head) and plane mirror (legs of normal size).

27. (a) Name the spherical mirror used as:
1. Shaving mirror,
 2. Rearview mirror in vehicles,
 3. Reflector in search – lights.
- (b) Write any three differences between a real and virtual image.

Ans : [CBSE 2012]

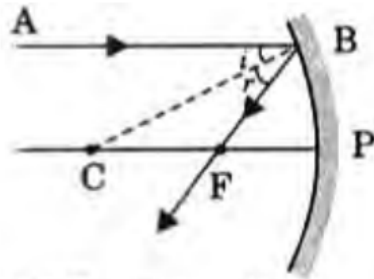
- (a)
1. Concave mirror,
 2. Convex mirror,
 3. Concave parabolic
- (b) Differences between a real and virtual image :

Real Image	Virtual Image
It can be taken on a screen	It can not be taken on a screen
It is always inverted	It is always erected
When reflected or refracted rays actually meet at a point, then real image is formed.	When reflected or refracted rays appear to meet at a point then virtual image is formed

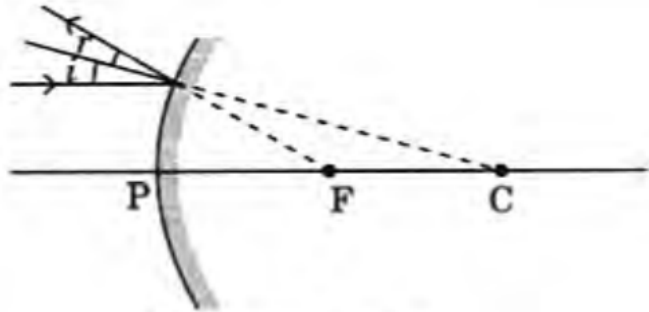
34. With the help of ray diagram show that angle of incidence is equal to the angle of reflection when a ray is incident on the concave/convex mirror.

Ans :

[CBSE 2014]



(a) Concave mirror



(b) Convex mirror

54. A student wants to project the image of a candle flame on a screen 60 cm in front of a mirror by keeping the flame at a distance of 15 cm from its pole.
- Write the type of mirror he should use.
 - Find the linear magnification of the image produced.
 - What is the distance between the object and its image?
 - Draw a ray diagram to show the image formation in this case.

Ans : [All India 2014]

Given: $v = -60$ cm

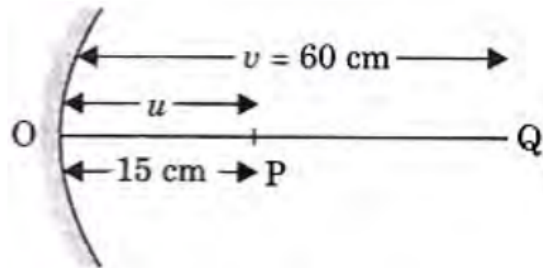
$u = -15$ cm

- a. Since the image is real the mirror is concave

b.
$$m = -\frac{v}{u} = \frac{-(-60)}{(-15)} = -4$$

$$m = -4$$

- c. Distance between object position and image position = $60 - 15 = 45$ cm



52. A 4.5 cm needle is placed 12 cm away from a convex mirror of focal length 15 cm. Give the location of image and magnification. Describe what happens to the image as the needle is moved farther from the mirror.

Ans : [All India 2014]

Given: $u = -12$ cm

$f = +15$ cm

$h_1 = 4.5$ cm

$v = ?, m = ?$

Mirror formula,

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{15} - \frac{1}{-12} = \frac{1}{15} + \frac{1}{12}$$

$v = +6.6$ cm

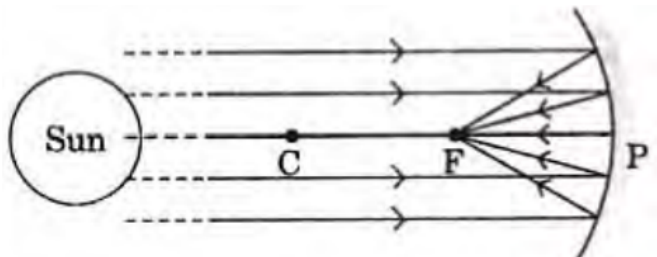
$$m = -\frac{v}{u} = -\frac{6.6}{-12} = 0.55$$

Size of image is smaller than size of object.

66. (a) Define real image of an object.
(b) Name the mirror that:
- can give real as well as virtual image of an object.
 - will always give virtual image of same size of an object.
 - will always give virtual and diminished image of an object.
 - is used by a doctor in examining teeth,
- (b) With the help of a ray diagram explain the use of concave mirror as solar concentrators.

Ans : [All India 2011]

- a. When the reflected rays after reflection actually meet at a point then real image is formed.
- b. Concave mirror
Plane mirror
Convex mirror
Concave mirror
- c. The rays coming from the sun are parallel to principal axis and will concentrate at focus after reflection.

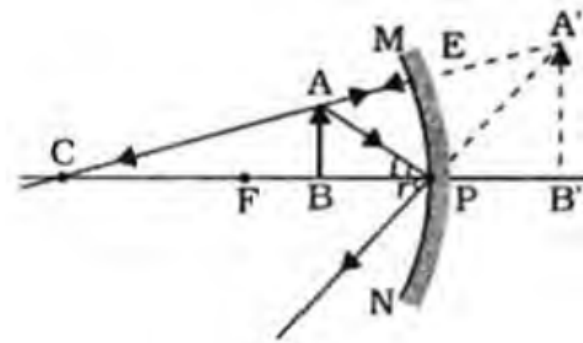


71. (a) Under what condition, a concave mirror produces a virtual and magnified image? Draw a labelled ray diagram to show the formation of image in the above case. Also, state the position of object to produce magnified and real image.
- (b) A ray of light moving along principal axis is falling on a concave mirror. Draw the path of reflected ray. Also, state the values of angle of incidence and reflection in this case.

Ans :

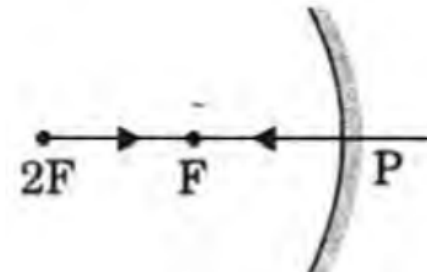
[All India 2011]

- a. A concave mirror will produce a virtual and magnified image of the object if placed in between F and pole of the mirror.



Real and magnified image will be formed if the object is placed in between $2F(C)$ and F in front of a concave mirror.

- b. Ray of light moving along the principal axis will retrace its path because incident angle $\angle i =$ reflected angle $\angle r = 0$.



- (b) (i) Give mathematical formulae for determining magnification produced by a spherical mirror.
- (ii) What does $m = -1$ signify? Identify the mirror that can produce it.

Ans :

[All India 2012]

84. State the relation between object distance, image distance and focal length of a concave or convex mirror. A concave mirror produces two times magnified real image of an object at 10 cm from it. Find the position of the image.

Ans :

[CBSE 2016]

For concave or convex mirrors the relation between u , v and f is given by

mirror formula, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$$m = -2$$

$$u = -10 \text{ cm}$$

$$m = -\frac{v}{u} = -2 \text{ or } v = 2u = -20 \text{ cm}$$

99. (a) A concave mirror produces three times enlarged image of an object placed at 10 cm in front of it. Calculate the focal length of the mirror.
- (b) Show the formation of the image with the help of a ray diagram when object is placed 6 cm away from the pole of a convex mirror.

Ans :

[All India 2012]

(a) $u = -10 \text{ cm}$

Let a real image is formed in concave mirror i.e.,

$$m = -\frac{v}{u} = -3$$

$$v = 3u$$

$$v = 3 \times 10 \text{ cm} = 30 \text{ cm}$$

Mirror formula,

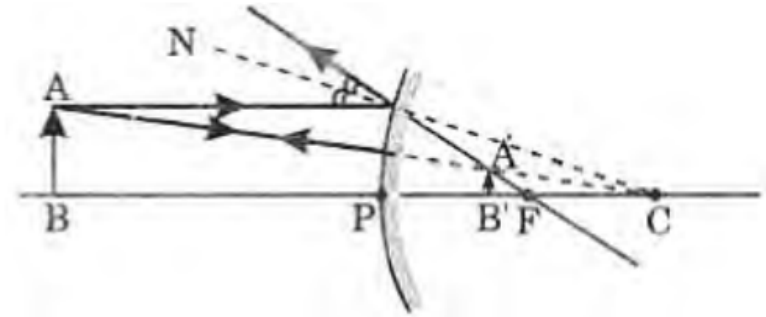
$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{-30} + \frac{1}{-10}$$

$$\frac{1}{f} = -\frac{1}{30} - \frac{1}{10} = \frac{-4}{30}$$

$$f = -\frac{30}{4} = -7.5 \text{ cm}$$

- (b) When $u = -6 \text{ cm}$ from pole of a convex mirror. For convex mirror, the image will be virtual, erected and smaller.



100. An object 1 cm tall is placed at a distance of 15 cm from a concave mirror of focal length 10 cm. Find the position, size and nature of the image formed.

Ans :

[All India 2013]

In concave mirror

$$h_1 = 1 \text{ cm}$$

$$u = -15 \text{ cm}$$

$$f = -10 \text{ cm}$$

$$v = ?$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} + \frac{1}{-15} = \frac{1}{-10}$$

$$\frac{1}{v} - \frac{1}{10} + \frac{1}{15} = -\frac{1}{30}$$

29. (a) A student has focussed the image of an object of height 3 cm on a white screen using a concave mirror of focal length 12 cm. If the distance of the object from the mirror is 18 cm, find the values of the following :

(i) Distance of the image from the mirror

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(ii) Height of the image

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29. A student wants to project the image of a candle flame on a screen 60 cm in front of a mirror by keeping the candle at a distance of 15 cm from its pole.

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(a) Name the type of mirror used.

(b) Also calculate :

(i) Magnification of the image produced

(ii) Distance between object and its image

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(c) Draw a ray diagram to show the image formation.

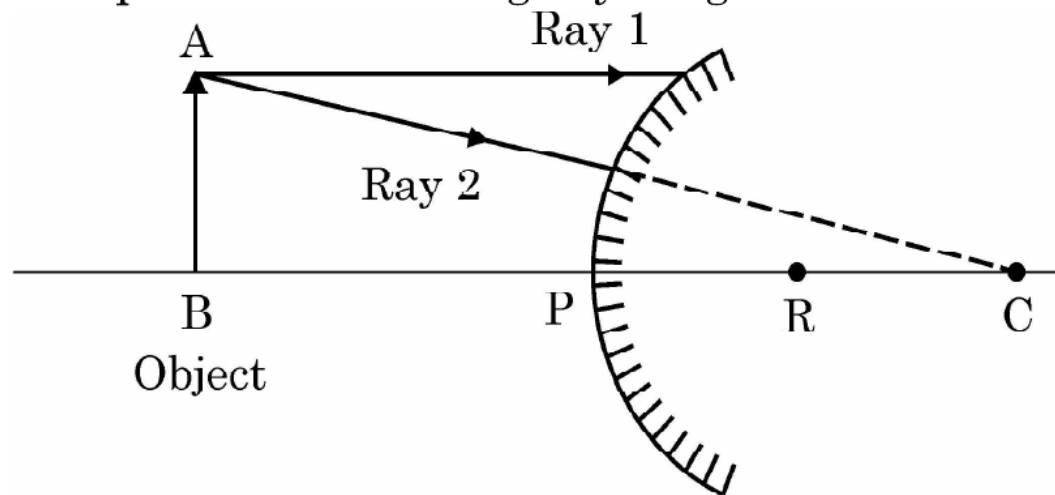
(c) An object 10 cm in size is placed at 100 cm in front of a concave mirror. If its image is formed at the same point where the object is located, find :

2

- (i) focal length of the mirror, and
- (ii) magnification of the image formed with sign as per Cartesian sign convention.

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30. (a) Complete the following ray diagram to show the formation of image : 3



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- (b) Mention the nature, position and size of the image formed in this case.
- (c) State the sign of the image distance in this case using the Cartesian sign convention.

29. The magnification produced when an object is placed at a distance of 20 cm from a spherical mirror is $+1/2$. Where should the object be placed to reduce the magnification to $+1/3$.

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26. (a) Find the magnification of the image formed by a spherical mirror from the following data :
 $u = -20\text{ cm}$, $f = -15\text{ cm}$.

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27. If we want to obtain a virtual and magnified image of an object by using a concave mirror of focal length 18 cm, where should the object be placed? Use mirror formula to determine the object distance for an image of magnification +2 produced by this mirror to justify your answer.

(iii) (B) An object is placed at a distance of 18 cm from the pole of a concave mirror of focal length 12 cm. Find the position of the image formed in this case.

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24. An object is placed at a distance of 10 cm from a convex mirror of focal length 15 cm. Find the position of the image formed by the mirror. CBSE 2024 **2**

31. (a) An object is placed in front of a concave mirror of focal length 12 cm. If distance of the object from the pole of the mirror is 8 cm, then use mirror formula to determine the position of the image formed. Draw a labelled ray diagram to justify your answer in this case. CBSE 2024 **3**

OR

31. (b) (i) The image of an object formed by a mirror is real, inverted and is of magnification-1. If the image is at a distance of 30 cm from the mirror, where is the object placed ? Give reason to justify your answer. **3**
- (ii) Where would the image be if the object is moved 15 cm towards the mirror ? Draw ray diagram for the new position of the object to justify your answer. CBSE 2024

(ii) A 1.5 cm tall candle flame is placed perpendicular to the principal axis of a concave mirror of focal length 12 cm. If the distance of the flame from the pole of the mirror is 18 cm, use mirror formula to determine the position and size of the image formed.

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- (c) (i) An object is placed at a distance of 10 cm from the pole of a convex mirror of focal length 15 cm. Find the position of the image.

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25. “The linear magnification produced by a spherical mirror is + 3.” Based on this statement answer the following questions : 2

- (a) What is the type of mirror ?
 - (b) Where is the object located ?
 - (c) List two properties of the image formed (other than the size/magnification).
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36. (a) (i) State laws of reflection of light.

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- (ii) An object of height 5.0 cm is placed at 15 cm in front of a concave mirror of focal length 10 cm. At what distance from the mirror should a screen be placed, so that a focussed image is obtained on it ? Find the height of the image.

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(c) (ii) A 3 cm long candle flame is placed in front of a concave mirror. If the distance between the pole and the candle flame is 40 cm and the image is formed at the same place where the candle flame is located, find the 2

(1) focal length of the mirror, and

(2) magnification of the image formed with sign as per the New Cartesian Sign Convention.

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